Clinical Study Summary Report (CSSR)
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1. Administrative & Study Identification

Full Study Title: An Open-label Study of AZD0120 in Adults With Multiple Sclerosis
Short Title/Acronym: ZENITH Trial
Protocol Number: D831DC00001
NCT Number: NCT07224373
Sponsor Name: AstraZeneca
Investigational Product: AZD0120 (BCMA/CD19 dual targeting CAR+ T-cell therapy)
Phase of Development: Phase Ib
Medical Monitor: AstraZeneca Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of AZD0120, a BCMA/CD19 dual targeting CAR+ T-cell therapy, in adult participants with Multiple Sclerosis to determine the Recommended Phase 2 dose (RP2D) for each disease cohort.
Secondary Objectives: To further characterize the efficacy, pharmacodynamic effects, immunogenicity, and changes in health outcomes of AZD0120.

2.2 Background & Rationale To address the need for highly effective therapies in patients with severe, active Multiple Sclerosis who have not responded well to standard disease-modifying treatments. AZD0120 targets the disease in a novel way by focusing on specific cells underlying the disease process. By utilizing a dual BCMA/CD19 CAR T-cell approach, this therapy aims to effectively target and reduce harmful immune cells, offering a new mechanism of action to manage symptoms, reduce inflammation, and potentially slow disease progression in both relapsing and progressive forms of Multiple Sclerosis.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional
Control Type: Single-arm
Blinding: Open-label
Randomization: N/A (Parallel assignment to two specific disease cohorts: Relapsing MS [RMS] and Progressive MS [PMS])

3.2 Planned Enrollment & Duration Target N: Approximately 24 participants (~9-12 participants evaluated per disease cohort).
Number of Sites: Multi-center (US, Canada, and Rest of World).

Estimated Study Start: 2024 / 2025
Estimated Primary Completion: TBD

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. [Age ≥ 18 to ≤ 75 years old at the time of consent]. 2. [Adequate physiological function and reserve at screening]. 3. [RMS Cohort: Diagnosis of RMS (2017 McDonald Criteria) or relapsing, active SPMS. EDSS of ≤ 6.5. Evidence of active disease or intolerance on high efficacy disease-modifying therapy for ≥ 6 months]. 4. [PMS Cohort: Diagnosis of PPMS or not active progressive SPMS. EDSS of ≥ 3.0 and ≤ 6.5. Inadequate response to ≥ 1 high-efficacy DMT with ≥ 6 months treatment or intolerance].

4.2 Exclusion Criteria 1. [Any prior CAR-T or CAR-NK cell exposure, prior autologous hematopoietic stem cell transplantation, or solid organ transplant]. 2. [Underwent splenectomy within 12 months prior to signing the ICF]. 3. [History of a seizure disorder, clinically significant psychiatric condition, or other immune-mediated disease requiring continued systemic immunosuppression]. 4. [Seropositive for HIV, active viral hepatitis (HBV, HCV), or pregnant/breastfeeding].

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: AZD0120 administered as a single dose following standard CAR-T lymphodepletion chemotherapy.
Route of Administration: Intravenous (IV) Infusion.
Duration of Treatment: Single infusion with long-term post-infusion follow-up and monitoring.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for Multiple Sclerosis symptom management.
Prohibited: Other high-efficacy disease-modifying therapies (heDMTs) for MS, systemic immunosuppressants, and live/live-attenuated vaccines during the acute monitoring period.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, MRI scans, EDSS evaluation, CSF exam/Lumbar Puncture, Safety Labs.
Baseline	Day 0	Apheresis, Lymphodepletion, AZD0120 Infusion
Interim	Weeks 1-4	Intensive inpatient/outpatient monitoring for Dose-Limiting Toxicities (DLTs), CRS, and Neurologic Toxicity (ICANS)
End of Study	Long-Term	Final Efficacy Assessments (MRI, relapses), Long-term CAR-T safety follow-up (up to 15 years)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of Dose-Limiting Toxicities (DLTs) and Treatment-Emergent Adverse Events (TEAEs) to determine the Recommended Phase 2 Dose (RP2D).
Measurement Method: Clinical assessment and Safety Review Committee (SRC) evaluation.

7.2 Secondary & Exploratory Endpoints [Pharmacodynamics and immunogenicity of AZD0120]
[Changes in health-related outcomes, including clinical relapses, EDSS changes, and MRI activity]

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive continuous monitoring for cellular therapy-related toxicities, including Cytokine Release Syndrome (CRS) and neurological toxicities.
Serious Adverse Events (SAEs): Reported within 24 hours of investigator awareness.
Data Monitoring Committee (DMC): A designated Safety Review Committee (SRC) is utilized to evaluate DLTs and determine the recommended dose.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive any fraction of the AZD0120 infusion).
Sample Size Justification: Approximately 9-12 participants per cohort is standard for a Phase Ib safety and dose-finding study to adequately assess DLTs and establish an RP2D profile.
Handling of Missing Data: Standard Phase I approaches will be utilized (e.g., observed cases for safety parameters).

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
Monitoring Plan: High-frequency onsite monitoring during the acute post-infusion period, transitioning to routine remote and onsite follow-ups.
Audit Trail: Validated eCRF systems to track patient safety, dose-limiting toxicities, and neurological assessments.

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11. Study Identifiers & Governance

Primary ID: {{D831DC00001}}
Registry Number: {{NCT07224373}}
Sponsor/Lead Organization: {{AstraZeneca}}
Study Title: {{ZENITH Trial}}
Official Regulatory Title: {{An Open-label Study of AZD0120 in Adults With Multiple Sclerosis}}

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12. Clinical Framework (Multiple Sclerosis Specific)

Primary Condition: {{Multiple Sclerosis (RMS and PMS)}}
Diagnosis Threshold: {{EDSS 3.0 to 6.5, 2017 McDonald Criteria}}
Medical Methodology: {{BCMA/CD19 dual targeting CAR+ T-cell therapy}}
Optimization Target: {{Safety, Tolerability, and RP2D determination}}

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13. Study Procedures & Methodology

Management Pathway: {{CAR-T clinical pathway (Apheresis, Lymphodepletion, Infusion)}}
Education Protocol (HCP): {{CRS and neurotoxicity management training}}
Education Protocol (Patient): {{CAR-T post-infusion symptom monitoring}}
Quality Audit Mechanism: {{Safety Review Committee (SRC) evaluation}}

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14. Observation & Follow-Up Schedule

Total Duration: {{Long-term follow-up (up to 15 years for CAR-T)}}
Onsite Visit Cadence: {{Intensive monitoring post-infusion, then regular intervals (e.g., Q4W to Q12W)}}
Remote Monitoring Frequency: {{Between onsite visits}}
Checklist Review Interval: {{Ongoing neurological and EDSS assessments}}

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]